Week 4 · Part E — justification

1 · Panels (question · why this chart · Week 2 rule)

P1 daily sessions — is the trend up once the weekend is removed? Line: position on a common scale, the best channel for a trend. Raw in light grey, signal in blue (one pop-out, grey for context). **P2 day×hour heat map** — when in the week does traffic live? A 7×24 grid has no ordering to exploit, so lightness; sequential Blues, dark = more. **P3 treemap** — where is the mass? Containment shows parent→child at a glance; area is the 4th-best channel, so every block prints its value and share; one hue per parent (cognitive load: four hues). **P4 sorted bars** — by how much? Length on a common zero baseline (rank 2) with values printed: the treemap answers *where*, the bars answer *how much*.

2 · Time decisions

Parse: two loggers — ISO (9,019 rows) and DD/MM/YYYY HH:MM (3,651). Ambiguity resolved by test: 2,151 slash rows have a first field above 12, none has a second field above 12, so it is DD/MM. Explicit formats; format='mixed' would silently misdate 1,382 rows. **Resolution/aggregation:** hourly → daily with sum(), because sessions are a flow (never last()); missing days stay NaN (13 days, gaps counted on the hourly index before resampling). **Window:** 7 days, centred, because the cycle removed is weekly (Part B.1: 31.7% of raw variation removed, 3 days lost each end); centred because this is a retrospective review — a live version would have to be trailing and ~3 days late. **Ethics clause:** the window was fixed at 7 days before plotting, for the weekly cycle. The series is shown from its first recorded day (1 Apr 2024) to its last (11 Sep 2025); no start date was selected. A 28-day version moves 16.0k → 30.4k against 16.0k → 30.0k for 7 days: the direction does not change. Part B.5 shows other start/window choices give 38 rising and 22 falling lines, so I report that range. Symmetry test: I would have fixed the same window had it weakened my message.

3 · Rows removed

Rule	Count	Effect
Exact duplicate hours (same timestamp, identical values)	286	12,670 → 12,384 rows
Sentinel sessions = −1 → missing (not zero); 287 hours imputed by weekday-hour median for daily sums only	287 (290 before dedupe)	daily total +2.39%
Days never recorded: left as gaps, not 0	13 days (312 h)	529 day slots, 516 observed
Partial final month (Sep 2025, 11 days): shaded, kept out of monthly comparisons	11 days	—
Duplicate SKU-0127 (also under Lifestyle/Wearables): dropped the wrong-parent copy	1 row	Wearables 10.1% → 9.3%; max share error 0.93 → 0.24 pp
Device spellings (6) folded to mobile/desktop/tablet	0 dropped	—
7 negative-net SKUs netted inside sub-categories, no negative area drawn	7 SKUs, −1.07 m	−0.06% of total

4 · Roll-ups

No *Other* bucket: four categories and eight sub-categories fit in the 4±1 chunks budget with nothing folded away (0 categories hidden, 0% of value). Depth cut at sub-category: the 345 SKUs sit below it and are not drawn (their negatives are netted in). Self-check: all eight shares are within 0.24 pp of shares_eight.csv. Synthetic course data, CDB601220 Week 4.